

- 3 lab excuses
- option C - only if USC campus support indicates an emergency occurred
- true
- I agree with the statement because I myself have noticed that I learn/retain so much more when I don't use AI for guidance. Sometimes it's hard to feel like I'm struggling, but it's ultimately worth it.
- false
- submit late up to the time of the next exam for 25% penalty
- can lead to errors
- can lead to errors
- cs104 - repos
- git add
- git log -3
- git clone git@github.com:usc-csci104-spring2026/resources.git

problem 3

part a

outer loop: $\sum_{i=2}^n \theta(1)$

$i = 2$

iteration 1: $i=2 \rightarrow 4 \rightarrow 2^2 \rightarrow 2^{2^1}$
 2: $i=4 \rightarrow 16 \rightarrow 2^4 \rightarrow 2^{2^2}$
 3: $i=8 \rightarrow 256 \rightarrow 2^8 \rightarrow 2^{2^3}$

loop continues until $2^{2^k} > n$

$$\log 2^{2^k} > \log n$$

$$2^k > \log n$$

$$\log 2^k > \log(\log n)$$

$$k > \log(\log n)$$

$$\sum_{i=2}^{\log(\log n)} \theta(1) = \theta(\log(\log n))$$

part b

outer loop: $\sum_{i=1}^n \theta(1)$

condition: $i \% \sqrt{n} == 0$

true whenever i is multiple of $\sqrt{n}$ from 1 to n

$i = \sqrt{n}, 2\sqrt{n}, 3\sqrt{n}, 4\sqrt{n} \dots \sqrt{n}\sqrt{n} \rightarrow$ coefficients: j

when triggered, inner loop: $i^2 = (j\sqrt{n})^2$
 condition & inner loop: $\sum_{j=1}^{\sqrt{n}} (j\sqrt{n})^3 = \sqrt{n}^3 \sum_{j=1}^{\sqrt{n}} j^3$

$$\sum_{j=1}^{\sqrt{n}} j^3 = \left(\frac{\sqrt{n}(\sqrt{n}+1)}{2} \right)^2 = \left(\frac{n+\sqrt{n}}{2} \right)^2 = (n)^2 = n^2$$

$$\sqrt{n}^3 (n^2) = n^{7/2} \cdot n^2 = n^{9/2} \cdot n^{1/2} = n^{7/2}$$

$$\theta(n^{7/2})$$

part c

outer loop: $\sum_{i=0}^n \theta(1)$
 middle loop: $\sum_{k=1}^i \theta(1)$

condition: if $A[k] == i$, runs until $A[k]3$ matches i

inner loop: $\sum_{m=1}^{2m} \theta(1)$

$m = 1, m = 2, 4, 8, 16 \dots$

$$2^0 \quad 2^1 \quad 2^2 \quad 2^3 \quad 2^4$$

$$2^{m-1} \quad 2^{m-1} \quad 2^{m-1} \quad 2^{m-1}$$

runs until $2^{m-1} > n$

$$\log 2^{m-1} > \log n$$

$$m-1 > \log n \rightarrow m > \log n$$

single comparison

given this, $A[k] == i$ only occurs once since i increments w/o repeating so it matches how many ever n times

$$n \cdot \theta(\log n) = \theta(n \log n)$$

$$\theta(n^2) + \theta(n \log n) = \theta(n^2)$$

part d

dynamic array, grows by $\frac{3}{2}$

outer loop: $a[i] = i \cdot i$, runs once per iteration

condition: resizes array - copies all size elements into new

size = 10, 15, 22, 33, 49, 73, 109

ratio $r = 1.5$ - geometric series

$$10 \left(\frac{3}{2} \right)^r$$

$$10 \left(\frac{3}{2} \right)^r \geq n$$

$$\left(\frac{3}{2} \right)^r \geq \frac{n}{10}$$

$$\log \left(\frac{3}{2} \right)^r \geq \log \left(\frac{n}{10} \right)$$

$$r \geq \log \left(\frac{n}{10} \right) \rightarrow \text{resizes}$$

copying the actual array into new array:

$$10 + 10 \left(\frac{3}{2} \right) + 10 \left(\frac{3}{2} \right)^2 + 10 \left(\frac{3}{2} \right)^3 \dots 10 \left(\frac{3}{2} \right)^r \text{ resizes}$$

$$10 \sum_{k=0}^r \left(\frac{3}{2} \right)^k = \frac{\left(\frac{3}{2} \right)^{r+1} - 1}{\frac{3}{2} - 1} = \frac{3}{2} - 1 = \frac{1}{2}$$

$$10 \left(\frac{\left(\frac{3}{2} \right)^{r+1} - 1}{\frac{3}{2} - 1} \right) = 20 \left(\left(\frac{3}{2} \right)^{r+1} - 1 \right) = 20 \left(\frac{3}{2} \right)^{r+1} - 20$$

$$\left(\frac{3}{2} \right)^r = \frac{n}{10} \rightarrow 20 \left(\frac{n}{10} - 1 \right) = \frac{20n}{10} - 20 = 2n$$

$$\theta(2n) = \theta(n)$$

4a

$in1 = 1, 2, 3, 4$

$in2 = 5, 6$

1. $llrec(1 \rightarrow 2 \rightarrow 3 \rightarrow 4, 5 \rightarrow 6)$

$1 \rightarrow next = llrec(5 \rightarrow 6, 2 \rightarrow 3 \rightarrow 4)$, return 1

2. $llrec(5 \rightarrow 6, 2 \rightarrow 3 \rightarrow 4)$

$5 \rightarrow next = llrec(2 \rightarrow 3 \rightarrow 4, 6)$, return 5

3. $llrec(2 \rightarrow 3 \rightarrow 4, 6)$

$2 \rightarrow next = llrec(6, 3 \rightarrow 4)$, return 2

4. $llrec(6, 3 \rightarrow 4)$

$6 \rightarrow next = llrec(3 \rightarrow 4, nullptr)$, return 6

5. $llrec(3 \rightarrow 4, null)$

return 3 (but 3 still points to 4)

$1 \rightarrow 5 \rightarrow 2 \rightarrow 6 \rightarrow 3 \rightarrow 4 \rightarrow null$

4c

$in1 = nullptr, in2 = 2$

2 is returned

